

32)  $Q = \int Idt$  The net amount of charge flowed pass a point is the area under the current time graph. Hence the maximum charge passing through the point is at time  $\frac{\pi}{\omega}$ . From time  $\frac{\pi}{\omega}$  to  $\frac{2\pi}{\omega}$ , there is a negative area under the graph denoting an amount of charge moving backwards through the same point. So the net charge is decreasing.  
 Answer A

32) This is a sine curve with a period of 0.2 s and amplitude of 0.5 V. Hence the equation